Sequences, sums and averages

L2-AL2 Level 2 Intermediate · Algebra

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Arithmetic sequences, Gauss pairing, means handled as totals, and working backwards through a process.
By the end you can count and add the terms of an arithmetic sequence without fence-post errors; turn any problem about changing means into a statement about totals; undo a multi-step process by working backwards.
Prerequisites: Thinking with algebra (L1-AL1). **Contests:** IMC, Intermediate Kangaroo, Cayley, Hamilton, AMC 10. **Time:** about 75 minutes.

1 Lesson

Twenty-five consecutive integers add up to 2025. What is the largest of them?

You could call the smallest one n and solve $n + (n+1) + \cdots + (n+24) = 2025$, and that works. But note that consecutive integers are evenly spaced, so their mean is the middle one. The mean is $2025 \div 25 = 81$, so the thirteenth number is 81, and the largest is twelve more, 93.

This unit treats sums and means through totals, pairing and the balance property of the mean. The unit also covers working backwards, for problems where the end of a process is known and the start is not.

1.1 Arithmetic sequences

An arithmetic sequence goes up (or down) by the same amount each time. That amount is the common difference d, and the sequence is $a,\ a+d,\ a+2d, \ldots$. To reach the nth term you take $n-1$ steps, not n.

Terms of an arithmetic sequence.
The nth term of the arithmetic sequence with first term a and common difference d is $a + (n-1)d$. Turned around, an arithmetic sequence that runs from a to l in steps of d has

$$\frac{l-a}{d} + 1$$

terms.

The $+1$ is the fence-post fact, since a fence with 10 gaps has 11 posts. For example, the numbers from 13 to 97 that leave remainder 1 when divided by 4 are $13, 17, \ldots, 97$, and there are $\frac{97-13}{4} + 1 = 22$ of them, not 21.

1.2 Gauss pairing

To add up an arithmetic sequence, write the sequence forwards and backwards and add the two rows column by column.

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Sum of an arithmetic sequence.
If an arithmetic sequence has n terms, first term a and last term l, then its sum is

$$S = \frac{n(a+l)}{2}.$$

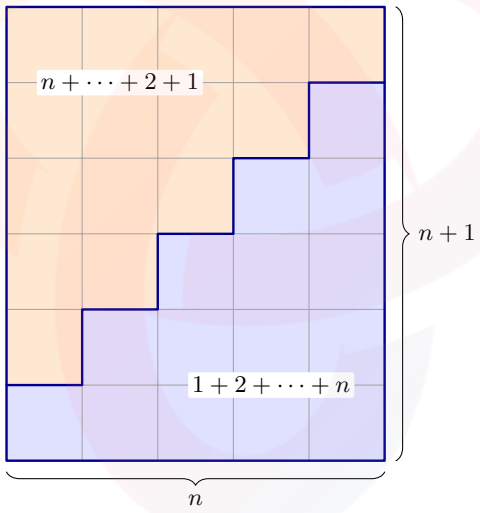
In words, the number of terms times the mean of the first and last. In particular, $1+2+\cdots+n = \frac{n(n+1)}{2}$.

Proof. Write the sum forwards and then backwards:

$$S = a + (a+d) + \cdots + (l-d) + l, \qquad S = l + (l-d) + \cdots + (a+d) + a.$$

In each column the two terms add to $a+l$, because one goes up by d exactly as the other goes down by d. There are n columns, so $2S = n(a+l)$. □

For $1+2+\cdots+n$ there is also a picture. Two copies of the staircase fit together into an n by $n+1$ rectangle, so each staircase contains $\frac{n(n+1)}{2}$ squares.



The formula is also a statement about means. The mean of an arithmetic sequence is $\frac{a+l}{2}$, which is the middle term when n is odd and halfway between the two middle terms when n is even. This is why the opening problem needed only one division. One more consequence is worth remembering, namely $1+3+5+\cdots+(2n-1) = \frac{n(1+2n-1)}{2} = n^2$.

1.3 Means and totals

The mean of a list is its total divided by the number of entries. Almost every problem about means becomes routine once you switch to totals, because totals add and means do not.

Working with means.

1. If n numbers have mean m, their total is nm.
2. If a number x is added to a list of n numbers with mean m, the new mean is

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$$\frac{nm+x}{n+1} = m + \frac{x-m}{n+1}.$$

3. (Balance.) The amounts by which the numbers differ from their mean add up to zero, counting amounts below the mean as negative.

The second fact is the first one rearranged, but note what the second fact says. Adding x moves the mean towards x by $\frac{1}{n+1}$ of the gap between them. The third is the first in another form, since $(x_1 - m) + \cdots + (x_n - m) = (x_1 + \cdots + x_n) - nm = 0$. So the mean is a balance point, and measuring from the mean is often the quickest method.

A common mistake is to forget that the mean of two groups put together is *not* the average of their two means, unless the groups are the same size. A group of 10 with mean 50 and a group of 30 with mean 70 have combined mean $\frac{500+2100}{40} = 65$, not 60.

1.4 Working backwards

Some problems describe a process step by step, tell you how the process ends and ask how the process started. Going forwards means carrying an unknown through every step, and the algebra grows at each step. Going backwards, you undo each step with known numbers.

The rule is to undo the steps in reverse order, replacing each operation by its inverse. If a step is "halve, then subtract 1", its undoing is "add 1, then double". The order matters, since "subtract 1, then double" gives a different answer.

Example 1.
How many terms does the sequence $7, 11, 15, \ldots, 403$ have, and what is their sum?

Solution.
The common difference is 4. A tempting answer for the number of terms is $\frac{403-7}{4} = 99$, but that counts the gaps between terms, not the terms. There are 99 steps of 4 from 7 to 403, so there are 100 terms. A check on a small case confirms this. From 7 to 15 there are $\frac{15-7}{4} = 2$ steps and 3 terms.
The sum is the number of terms times the mean of the first and last, which is $100 \times \frac{7+403}{2} = 100 \times 205 = 20500$.
Pairing directly, $7 + 403 = 11 + 399 = 15 + 395 = 410$, and the 100 terms make 50 such pairs, giving $50 \times 410 = 20500$ again.

Example 2.
A class sat a test and their mean mark was 62. Four students who were absent sat the test later. Their mean mark was 74, and once they were included the mean for the whole class rose to 64. How many students sat the test on the original day?

Solution.
We cannot find individual marks, and we do not need them. Let n be the number of students on the original day, and convert everything to totals. The original total is $62n$, the latecomers add $4 \times 74 = 296$, and the final total is $64(n+4)$. So $62n + 296 = 64n + 256$, which gives $2n = 40$ and $n = 20$.
The balance property gives a mental method. Measure everything from the final mean of 64. The n original students are on average 2 below the final mean, so together they are $2n$ below.

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The four latecomers are on average 10 above the final mean, so together they are 40 above. Differences from the mean add up to zero, so $2n = 40$ and $n = 20$.
Note that we measured from the final mean, because the balance property applies to the mean of the combined list.

Example 3.
A pile of counters is changed three times. Each time, one third of the counters in the pile are removed and then 4 new counters are added. After the third change there are 20 counters. How many were there at the start?

Solution.
Going forwards with x counters at the start, after one change there are $\frac{2}{3}x + 4$. After two there are $\frac{2}{3}\left(\frac{2}{3}x + 4\right) + 4 = \frac{4}{9}x + \frac{20}{3}$, and after three there are $\frac{8}{27}x + \frac{76}{9}$. Setting this equal to 20 gives $\frac{8}{27}x = \frac{104}{9}$, so $x = 39$. This works, but the fractions are easy to get wrong.
Working backwards is shorter. One change does "multiply by $\frac{2}{3}$, then add 4". To undo this change, subtract 4 and then multiply by $\frac{3}{2}$. Starting from the end, $20 \to 16 \to 24$, then $24 \to 20 \to 30$, then $30 \to 26 \to 39$. So there were 39 counters at the start.
Always run the answer forwards as a check. Here $39 \to 26 \to 30$, then $30 \to 20 \to 24$, then $24 \to 16 \to 20$. The check also confirms that a third of the pile was a whole number each time, which the question implicitly requires.

Example 4.
In how many ways can 105 be written as a sum of two or more consecutive positive integers?

Solution.
Trial and error finds a few, namely $52 + 53$, then $34 + 35 + 36$, then $19 + 20 + 21 + 22 + 23$. To be sure of the count we need a systematic method. Suppose the sum has k terms starting at a, so the sum is $a + (a+1) + \cdots + (a+k-1)$. By Gauss pairing this equals $\frac{k(2a+k-1)}{2}$, so we need

$$k(2a + k - 1) = 210.$$

Call the second factor $m = 2a + k - 1$. Note two facts. First, $m - k = 2a - 1$ is odd, so one of k and m is even and the other is odd. Second, $a \geq 1$ means that m is bigger than k. Conversely, any factorisation $210 = k \times m$ with k smaller than m and $m - k$ odd gives a positive whole number $a = \frac{m-k+1}{2}$, and so a valid sum.
Since $210 = 2 \times 105$ with 105 odd, in every factorisation one factor is even and the other is odd, so the parity condition is automatic here. The factor pairs with the smaller factor first are 1×210, 2×105, 3×70, 5×42, 6×35, 7×30, 10×21 and 14×15. That makes eight, but $k = 1$ is the one-term "sum" 105 itself, which is excluded. So there are 7 ways. The last pair, $k = 14$ and $m = 15$, gives $a = 1$, and indeed $1 + 2 + \cdots + 14 = 105$.
The eight factor pairs match the eight odd divisors of 105. In general, the same argument shows that the number of ways to write N as a sum of one or more consecutive positive integers equals the number of odd divisors of N. In particular, the powers of 2 are exactly the numbers that cannot be written as a sum of two or more consecutive positive integers.

Common traps.

- The fence-post error. From a to l in steps of d there are $\frac{l-a}{d} + 1$ terms, not $\frac{l-a}{d}$.

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- Averaging averages. Combine groups through their totals, never by taking the mean of the means.
- Undoing steps in the forward order. The undoing of "add 3, then double" is "halve, then subtract 3", not "subtract 3, then halve".
- Not reading what kind of numbers are allowed. "Consecutive integers" and "consecutive positive integers" are different questions, because negative integers and zero can cancel each other out.
- Assuming the middle of an arithmetic sequence with an even number of terms is one of the terms. With an even count, the mean sits halfway between the two middle terms.

1.5 How to spot this in a contest

The triggers are evenly spaced numbers, the word "consecutive", phrases like "each time the value increases by", and any question where a mean changes. With means, write down totals straight away. With consecutive numbers, think of the middle term, because the sum is the number of terms times the middle. When a problem describes a process and tells you the result, ask whether each step can be undone, and if so, run the step backwards. When a sum looks too long to add, look for the pairing, which is first with last, second with second last, and so on.

Many IMC and AMC 10 questions on this material take under a minute once you see them this way, which leaves more time for the hard questions at the end of the paper.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ What is the sum of all the multiples of 3 between 100 and 200?

Problem 2 ⋆⋆ Six friends share a bag of sweets. The first friend takes half of the sweets in the bag and then one more. The second friend then takes half of the sweets that remain and then one more, and so on. Each friend in turn takes half of what is left and then one more. After the sixth friend has had a turn, the bag is empty. How many sweets did the third friend take?

Problem 3 ⋆⋆ A list of n numbers has the following property. Removing its largest number lowers the mean by 2, while removing its smallest number raises the mean by 3. The largest and smallest numbers differ by 75. Find n.

Problem 4 ⋆⋆⋆ How many of the integers $1, 2, 3, \ldots, 99$ can be written as a sum of *three or more* consecutive positive integers?

Problem 5 ⋆⋆⋆ An arithmetic sequence has an odd number of terms. The terms in the odd positions (first, third, fifth, and so on) add up to 520, and the terms in the even positions add up to 480. How many terms does the sequence have?

Problem 6 ⋆⋆⋆⋆ How many increasing arithmetic sequences of positive integers, with at least three terms, have terms that add up to 100?

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3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 10A 2021 Problem 4** ⋆
   Number of terms times the mean of the first and last
2. **AMC 10A 2020 Problem 8** ⋆
   Group the terms in blocks of four
3. **UKMT IMC 2016 Question 13** ⋆⋆
   Add up all the line totals and count how often each circle is used
4. **AMC 10A 2016 Problem 8** ⋆⋆
   Start from the end and undo each crossing
5. **UKMT IMC 2017 Question 7** ⋆⋆
   Turn the mean into a total
6. **AMC 10A 2021 Problem 5** ⋆⋆
   Work with totals, not means
7. **AMC 10A 2019 Problem 5** ⋆⋆
   Read carefully what kind of integers are allowed
8. **AMC 10B 2019 Problem 13** ⋆⋆⋆
   Consider each possible position of x in the ordered list
9. **UKMT IMC 2018 Question 24** ⋆⋆⋆
   Turn the mean into a total and see what the median and mode force
10. **UKMT Cayley 2018 Question C5** ⋆⋆⋆
    How much does changing one sign reduce the total?
11. **AMC 10A 2019 Problem 15** ⋆⋆⋆
    Look at the reciprocals of the terms
12. **AMC 10A 2021 Problem 22** ⋆⋆⋆⋆
    The missing pages are consecutive numbers. Compare totals

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4 Hints

1. Find the first and last multiples, count the terms carefully, then pair them up.
2. Undo one turn. If r sweets are left after a friend's turn, how many were there before that turn?
3. Let the mean be m and write each removal as an equation between totals. Express the largest and smallest numbers in terms of m and n.
4. If there are k terms starting at a, then $k(2a+k-1)=2N$. One factor is odd and the other even. Which numbers N have no such factorisation with $k \geq 3$?
5. The terms in odd positions form an arithmetic sequence of their own, and so do the terms in even positions. What is the mean of each?
6. With n terms, first term a and difference d, the condition is $n(2a+(n-1)d)=200$. First show that n cannot be large, then deal with each possible n separately.

5 Solutions

Problem 1

What is the sum of all the multiples of 3 *between* 100 *and* 200*?*

The smallest multiple of 3 above 100 is 102 and the largest below 200 is 198. They form an arithmetic sequence with common difference 3, so there are $\frac{198-102}{3}+1=33$ terms. The sum is the number of terms times the mean of the first and last, which is $33 \times \frac{102+198}{2} = 33 \times 150 = 4950$.

Answer: 4950

Problem 2

Six friends share a bag of sweets. The first friend takes half of the sweets in the bag and then one more. The second friend then takes half of the sweets that remain and then one more, and so on. Each friend in turn takes half of what is left and then one more. After the sixth friend has had a turn, the bag is empty. How many sweets did the third friend take?

If a friend finds y sweets, they take $\frac{y}{2}+1$ and leave $\frac{y}{2}-1$. So if r sweets are left after a turn, there were $2(r+1)$ before that turn. Working backwards from the empty bag, there were $2(0+1)=2$ sweets before the sixth turn, $2(2+1)=6$ before the fifth, $2(6+1)=14$ before the fourth and $2(14+1)=30$ before the third. The third friend found 30 sweets and took $15+1=16$. As a check, the bag started with $2(30+1)=62$ before the second turn and $2(62+1)=126$ at the very start, and running forwards $126 \to 62 \to 30 \to 14 \to 6 \to 2 \to 0$ works.

Answer: 16

Problem 3

A list of n numbers has the following property. Removing its largest number lowers the mean by 2*, while removing its smallest number raises the mean by* 3*. The largest and smallest numbers differ by* 75*. Find n.*

Let the mean be m, so the total is nm. Let the largest number be L and the smallest be S. Removing L leaves $n-1$ numbers with mean $m-2$, so $nm-L=(n-1)(m-2)$, which simplifies to $L=$

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$m + 2(n-1)$. Removing S leaves $n-1$ numbers with mean $m+3$, so $nm - S = (n-1)(m+3)$, which simplifies to $S = m - 3(n-1)$. Subtracting, $L - S = 5(n-1) = 75$, so $n = 16$.

Answer: 16

Problem 4

How many of the integers $1, 2, 3, \ldots, 99$ *can be written as a sum of* three or more *consecutive positive integers?*

If there are k terms starting at a, Gauss pairing gives $\frac{k(2a+k-1)}{2} = N$, so $k \times m = 2N$ where $m = 2a + k - 1$. Here $m - k = 2a - 1$ is odd, so one of k, m is odd and the other even, and $a \geq 1$ means m is bigger than k. Conversely any such factorisation of $2N$ gives a positive integer $a = \frac{m-k+1}{2}$. So N is a sum of three or more consecutive positive integers exactly when $2N$ can be written as (odd)×(even) with the smaller factor at least 3. Finding the numbers that fail is easier.

Write $N = 2^j q$ with q odd. In a factorisation of $2N = 2^{j+1}q$ into an odd and an even factor, the even factor contains all of 2^{j+1}. If $q = 1$, the only odd factor is 1, so the smaller factor is 1 and N fails. These are the powers of 2. If q is greater than 1, take the odd factor q and the even factor 2^{j+1}. They are different, and the smaller of them is at least 3 unless the factor is $2^{j+1} = 2$, which happens only when $j = 0$. So every N with q greater than 1 and $j \geq 1$ works (for example $10 = 1 + 2 + 3 + 4$).

That leaves odd N greater than 1, where $2N = 2 \times N$. If N has an odd divisor d with 1 less than d less than N, then $2N = d \times \frac{2N}{d}$ has odd factor $d \geq 3$ and even factor $\frac{2N}{d} \geq 6$, so N works (for example $9 = 2 + 3 + 4$). If N is an odd prime, the only factorisations are $1 \times 2N$ and $2 \times N$, so N fails, because N is a sum of two consecutive integers but not of three or more.

So the numbers that fail are the powers of 2 (that is $1, 2, 4, 8, 16, 32, 64$, seven of them) and the odd primes below 100 (there are 24 of them, since 25 primes lie below 100 and 2 is already counted). The answer is $99 - 7 - 24 = 68$.

Answer: 68

Problem 5

An arithmetic sequence has an odd number of terms. The terms in the odd positions (first, third, fifth, and so on) add up to 520*, and the terms in the even positions add up to* 480*. How many terms does the sequence have?*

Say there are $2n + 1$ terms, and let M be the middle term, in position $n + 1$. The $n + 1$ terms in odd positions form an arithmetic sequence whose first and last terms are the first and last terms of the whole sequence, so their mean is the mean of those two, which is M. The n terms in even positions also form an arithmetic sequence, with first and last terms in positions 2 and $2n$, which are symmetric about the middle, so their mean is also M. Hence $(n+1)M = 520$ and $nM = 480$. Subtracting gives $M = 40$, so $n = 12$ and the sequence has $2n + 1 = 25$ terms.

Answer: 25

Problem 6

How many increasing arithmetic sequences of positive integers, with at least three terms, have terms that add up to 100*?*

Let the sequence have $n \geq 3$ terms, first term $a \geq 1$ and common difference $d \geq 1$. Its sum is $\frac{n(2a+(n-1)d)}{2}$, so we need $n \times m = 200$ where $m = 2a + (n-1)d$. In particular n divides 200. Also $m \geq 2 + (n-1) = n + 1$, so $n(n+1) \leq 200$, which forces $n \leq 13$. The divisors of 200 from 3 to 13

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are 4, 5, 8 and 10.

If $n = 4$, then $2a + 3d = 50$, so d is even, and $a = \frac{50-3d}{2} \geq 1$ needs $d \leq 16$. That gives $d = 2, 4, \ldots, 16$, which is 8 sequences. If $n = 5$, then $2a + 4d = 40$, so $a = 20 - 2d \geq 1$, giving $d = 1, 2, \ldots, 9$, which is 9 sequences. If $n = 8$, then $2a + 7d = 25$, so d is odd, and $d = 1$ gives $a = 9$, $d = 3$ gives $a = 2$, while $d \geq 5$ makes a negative, giving 2 sequences. If $n = 10$, then $2a + 9d = 20$, so d is even, and only $d = 2$, $a = 1$ works, giving 1 sequence. The total is $8 + 9 + 2 + 1 = 20$.

Answer: 20